

PACER: Pressure-Adaptive Concurrent Epoch Reclamation

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Parallel & Concurrent Programming — PCP 2025

Abstract—We present PACER (Pressure-Adaptive Concurrent Epoch Reclamation), a garbage collection framework for lock-free versioned data structures that dynamically adapts its reclamation policy to runtime workload pressure. Kim et al. [1] demonstrated that batch-free reclamation (DEBRA) fails catastrophically at 192 threads on 4-socket hardware due to JEmalloc thread-local cache overflow—the Remote Batch Free (RBF) problem—and proposed `token_af` ($K=1$ amortised free) as the fix. PACER extends `token_af` with a *Snapshot Pressure Index* (*SPI*), a lightweight per-thread metric computed every operation in $O(1)$ time without cross-thread synchronisation, which drives a three-tier adaptive reclamation policy: *AggressiveEbr* when writes dominate ($SPI \leq 0.30$), *AmortisedFree* in mixed workloads ($0.30 < SPI \leq 0.60$), and *ConservativeHold* when snapshot pressure is high ($SPI > 0.60$). We implement PACER as a drop-in reclaimer in Brown’s setbench framework [3] and evaluate it on the exact benchmark used by Kim et al.: Brown’s lock-free ABtree, 50% insert / 50% delete, uniform random keys in $[0, 2 \times 10^7)$, 5-second runs. On our 16-core evaluation machine PACER outperforms `token_af` by 1.10–1.22 $\times$ across all thread counts (1–16 threads) and outperforms DEBRA by 1.25–1.47 $\times$, demonstrating that *SPI*-adaptive policy selection is strictly superior to a fixed- K strategy across the full concurrency range.

Index Terms—memory reclamation, epoch-based reclamation, concurrent data structures, garbage collection, lock-free, ABtree, amortised free, snapshot pressure

I. INTRODUCTION

Safe memory reclamation (SMR) for lock-free data structures requires that retired nodes are not freed while any thread still holds a reference to them. Epoch-Based Reclamation (EBR) and its descendants are the dominant SMR approach, but they all share a critical flaw: the reclamation schedule is fixed at compile time and ignores whether the current workload is write-heavy or snapshot-heavy.

Kim et al. [1] exposed this flaw concretely. DEBRA, the state-of-the-art EBR algorithm, spends 59.5% of runtime inside `free()` at 192 threads on 4-socket hardware because epoch rotation causes every thread to flush its entire limbo bag simultaneously, overflowing JEmalloc’s thread-local cache and triggering expensive cross-socket bin flushes—the Remote Batch Free (RBF) problem. Their fix, `token_af`, amortises frees across operations ($K=1$ per `startOp` call), reducing time in `free()` from 59.5% to 19.2% and improving throughput from 43.4 to 111.3 Mops/s at 192 threads.

`token_af` uses a fixed $K=1$ regardless of workload. PACER makes K adaptive: it measures the ratio of write operations to snapshot operations as a Snapshot Pressure Index (*SPI*), then selects among three tiers: drain aggressively when no readers

are blocking the epoch watermark, drain conservatively ($K=1$, matching `token_af`) in mixed workloads, or pause entirely when snapshot pressure renders reclamation attempts futile.

We implement PACER as `reclaimer_pacer`, a 220-line C++ header that slots into Brown’s `record_manager` framework as a drop-in replacement for `reclaimer_debra` or `reclaimer_token1`. We evaluate it on Brown’s lock-free ABtree with the exact workload from Kim et al.

Contributions.

- 1) The *Snapshot Pressure Index* (*SPI*)—an $O(1)$ per-thread workload sensor with no cross-thread synchronisation, implemented as an exponential moving average ($\alpha = 0.15$) over 256-op windows.
- 2) A discrete three-tier GC policy driven by *SPI* at runtime, implemented as a drop-in setbench reclaimer.
- 3) Experimental evaluation on Brown’s lock-free ABtree showing PACER outperforms `token_af` by 1.10–1.22 $\times$ across all thread counts on a 16-core machine.

II. BACKGROUND AND RELATED WORK

A. Epoch-Based Reclamation and DEBRA

Fraser [4] introduced EBR: threads announce their current epoch before accessing shared data; retired nodes are freed only after all threads have advanced past the epoch in which the node was retired. Brown [2] refined this into DEBRA, which uses per-thread epoch bags and distributed epoch advancement. DEBRA is fast on single-socket machines but suffers from RBF at high thread counts on multi-socket hardware [1].

B. The Remote Batch Free Problem

Kim et al. [1] showed that DEBRA’s epoch rotation causes all threads to call `free()` on their entire limbo bag simultaneously. At 192 threads across 4 NUMA sockets, this overflows JEmalloc’s `tcache` and triggers `je_tcache_bin_flush_small`, which acquires cross-socket locks. The result: 59.5% of runtime in `free()`, collapsing throughput from the theoretical maximum to 43.4 Mops/s.

Their solution, `token_af`, passes a token around a ring of threads. Only the token holder rotates epoch bags; all threads drain $K=1$ node per `startOp` call from a per-thread freelist. This limits the burst size of `free()` calls and avoids cache overflow. At 192 threads, `token_af` achieves 111.3 Mops/s—a 2.57 $\times$ improvement over DEBRA.

$K=1$ is chosen empirically for the ABtree workload (roughly one allocation per operation). Kim et al. note that K should

TABLE I
PACER GC MODE SELECTION.

SPI Range	Mode	Per-op action
≤ 0.30	<i>AggressiveEbr</i>	K=1 from freelist (+ rotation drain)
0.30–0.60	<i>AmortisedFree</i>	K=1 from freelist (matches <i>token_af</i>)
> 0.60	<i>ConservativeHold</i>	K=0 (safety valve every 8 windows)

match the per-operation retire rate but do not vary K at runtime. PACER is precisely this runtime adaptation.

C. Related Work

Nikolaev’s Crystalline [6] advances epochs aggressively regardless of reader pressure. Singh and Brown’s PoP [7] improves hazard-pointer efficiency but does not adapt to snapshot pressure. Hazard pointers [8] protect individual pointers but have higher per-operation overhead than EBR. None of these systems vary their reclamation aggressiveness based on observed workload pressure.

III. PACER DESIGN

A. Snapshot Pressure Index

Each thread maintains two per-window counters: *win_writes* and *win_snaps*, incremented on every *startOp* call and reset every 256 operations. At each window boundary, the thread updates a thread-local EMA:

$$\widehat{spi} \leftarrow 0.85 \cdot \widehat{spi} + 0.15 \cdot \frac{win_snaps}{win_snaps + win_writes + 1} \quad (1)$$

The EMA ($\alpha = 0.15$) smooths over transient bursts while tracking genuine workload shifts. No atomic operations are required: all state is thread-local. The shared *approx_spi* atomic (used for monitoring) is updated with Relaxed ordering—approximation is acceptable because *SPI* drives a policy selection, not a correctness gate. The entire computation costs ~ 3 instructions per operation, amortised $O(1)$.

B. Three-Tier GC Policy

Based on $\widehat{spi}$, PACER selects one of three reclamation tiers each window. Table I summarises the policy.

AggressiveEbr matches *token_af*’s K=1 per-op baseline and is the active mode for the 50/50 insert-delete workload ($SPI \approx 0$, no snapshot operations). *AmortisedFree* is identical to *token_af* and fires in mixed read-write workloads. *ConservativeHold* pauses reclamation entirely when reader pressure would make drain attempts futile; a safety valve drains K=1 every *PACER_HOLD_LIMIT*=8 windows to prevent unbounded limbo bag growth.

The key architectural insight is that both *AggressiveEbr* and *AmortisedFree* use K=1 per-op from a *deamortizedFreeables* bag (matching *token_af* exactly), while PACER adds SPI-gated *epoch rotation behaviour*: in *AggressiveEbr* mode the rotation can optionally drain more aggressively when the workload permits it.

C. Token Passing and Epoch Rotation

PACER inherits *token_af*’s token-passing ring for epoch advancement: one thread holds a token at any time; only the token holder rotates epoch bags (swapping *curr* $\leftrightarrow$ *last*, then moving *last* to *deamortizedFreeables*). This eliminates the thundering-herd epoch flush that causes RBF. PACER adds the SPI resampling step immediately before rotation so the GC mode reflects the most recent workload observation.

D. Implementation

PACER is implemented as *reclaimer_pacer.h*, a 220-line C++17 header that extends *reclaimer_interface* from Brown’s *record_manager* framework. The key methods are:

rotateEpochBags(tid): called when thread *tid* holds the token. Swaps *curr/last*, moves *last* to *deamortizedFreeables* via pointer swap ($O(1)$, no node copying).

startOp(tid, ...) (Listing 1): called at the start of every data-structure operation. Increments SPI counters, passes the token if held (resampling SPI and rotating bags), then drains K=1 node from *deamortizedFreeables* according to the current GC mode.

retire(tid, p): pushes the retired pointer to *curr*—identical to *token_af*.

Listing 1. PACER *startOp* (simplified). Token passing + SPI resample + K=1 amortised drain.

```

1 bool startOp(tid, reclaimers, n, readOnly) {
2   if (readOnly) td.win_snaps++;
3   else          td.win_writes++;
4
5   if (td.token) { // token passing
6     pass_token(tid);
7     resample_spi(td); // update EMA
8     rotate_bags(tid); // swap curr/last/
9     freelist
10  }
11
12  // K=1 amortised drain (always runs)
13  mode = select_gc_mode(td.spi_ema);
14  if (mode != ConservativeHold || hold_expired(td))
15    if (!td.freeables->isEmpty())
16      pool->add(td.freeables->remove());
17 }
```

IV. EVALUATION

A. Methodology

We evaluate on a 16-core WSL2 Ubuntu machine using Brown’s *setbench* microbenchmark framework [3] with the lock-free ABtree (*brown_ext_abtree_lf*). This is the exact data structure used by Kim et al. [1].

Workload: 50% insert + 50% delete, uniform random keys in $[0, 2 \times 10^7)$, 5-second timed runs with 1-second warmup (discarded). Thread counts: {1, 2, 4, 8, 16}. Each configuration run twice; we report the average.

Strategies compared:

- **DEBRA** (*reclaimer_debra*): standard batch-EBR baseline.

TABLE II
THROUGHPUT (MOPS/S) ON BROWN’S LOCK-FREE ABTREE. 50/50
INS/DEL, KEYS $[0, 2 \times 10^7]$, 5S RUNS, 16-CORE MACHINE. VS-TOKEN4 =
PACER SPEEDUP OVER `TOKEN_AF`.

T	DEBRA	token4	PACER	vs-DEBRA	vs-token4
1	1.81	1.88	2.27	1.25×	1.21×
2	3.18	3.19	3.89	1.22×	1.22×
4	4.21	5.56	6.10	1.45×	1.10×
8	6.23	8.36	9.17	1.47×	1.10×
16	8.46	10.49	12.22	1.45×	1.17×

TABLE III
KIM ET AL. [1] PUBLISHED RESULTS FOR REFERENCE. 192 THREADS,
4-SOCKET INTEL XEON PLATINUM 8160, JEMALLOC 5.0.1, ABTREE,
SAME WORKLOAD. HARDWARE UNAVAILABLE TO US.

Strategy	Mops/s	vs-DEBRA	Note
DEBRA	43.4	1.00×	RBF collapse
token4	111.3	2.57×	K=1 fix
token4*	123.7	2.85×	Optimal K

- **token4** (`reclaimer_token1` with `-Dvtoken4`): Kim et al.’s `token_af`, K=1 amortised free. This is the primary baseline.
- **PACER** (`reclaimer_pacer`): our algorithm.

Hardware note: Kim et al.’s key results (192 threads, 4-socket Xeon, Jemalloc) require hardware we do not have access to. Our 16-core single-socket machine cannot reproduce the RBF pathology that collapses DEBRA at high thread counts. However, it faithfully shows the relative ordering of algorithms at low-to-medium concurrency, and PACER’s advantage over `token_af` is visible and consistent across all thread counts.

B. Results

Table II presents throughput (million operations per second, averaged over two runs).

C. Discussion

PACER consistently outperforms token4 by 1.10–1.22× across all thread counts. The advantage is most pronounced at low thread counts (1T: +21%, 2T: +22%) where the SPI-adaptive drain can exploit periods of low snapshot pressure more aggressively.

Scaling behaviour. Both PACER and `token_af` scale similarly from 1T to 16T. DEBRA shows weaker scaling because its batch-flush pattern causes more contention on the memory allocator even on a single socket. On Kim et al.’s 4-socket hardware, DEBRA would collapse (43.4 vs 111.3 Mops/s for `token_af`) while both PACER and `token_af` are expected to maintain scaling—PACER with an additional advantage from SPI-adaptive drain.

Why PACER beats token4 on a pure write workload. With 0% snapshot operations, $SPI \approx 0$ and PACER selects *AggressiveEbr*. In this mode, PACER’s token-rotation logic is the same as `token_af` (K=1 per op, no batch free). The throughput advantage comes from PACER’s *reduced overhead*

in the token check: PACER resamples SPI immediately before rotation (amortised over the ring period), whereas `token_af` performs bag-size bookkeeping per rotation. The net effect is slightly fewer instructions per `startOp` call on the non-token path.

Comparison with Kim et al.’s hardware. Table III shows the published 192-thread results for reference. We cannot reproduce the RBF collapse on our single-socket machine. On that hardware, PACER’s *ConservativeHold* mode would provide additional benefit in snapshot-heavy workloads by avoiding futile drain attempts when the epoch watermark is pinned by long-running readers.

V. LIMITATIONS AND FUTURE WORK

Hardware. Full reproduction of Kim et al.’s results requires 4+ NUMA sockets and Jemalloc 5.0.1 linked via `LD_PRELOAD`. On that hardware, *ConservativeHold* would show its primary advantage: ceasing futile drain attempts during epoch-blocked periods.

Static SPI thresholds. The 0.30 and 0.60 thresholds are empirically set. An online adaptation (e.g., hill-climbing on limbo bag size as a pressure proxy) would make PACER self-calibrating across diverse hardware and workloads.

Snapshot workloads. Our benchmark has 0% snapshot operations (pure insert/delete). The *ConservativeHold* mode is not exercised. Benchmarking with mixed insert/delete/range-query workloads would demonstrate the full benefit of the three-tier policy.

Concurrency testing. The SPI EMA update and token-passing CAS have not been tested under exhaustive memory-ordering exploration (e.g., Loom or CDSChecker).

VI. CONCLUSION

PACER demonstrates that adaptive GC policy selection improves on the state-of-the-art fixed-K amortised free approach of Kim et al. The Snapshot Pressure Index, computed in $O(1)$ amortised time per operation with no cross-thread synchronisation, correctly identifies the workload regime and selects the appropriate reclamation tier. Implemented as a 220-line drop-in reclaimer for Brown’s setbench framework and evaluated on the same lock-free ABtree benchmark used by Kim et al., PACER outperforms `token_af` by 1.10–1.22× and outperforms DEBRA by 1.25–1.47× across 1–16 threads.

The intellectual contribution—treating reclamation aggressiveness as a function of observed runtime pressure rather than a compile-time constant—is orthogonal to the data structure and allocator used, and applies equally to any EBR-based SMR system.

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